Gokturk Colak

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Education

Istanbul University — Beyazit, Istanbul
Bachelor's in Economics, English Program — *August 2022 – Present*

- **GPA: 3.70/4.00**
- **Relevant Coursework:** Econometrics, Statistics, Mathematical Economics, Data Structures (Python), Data Analysis (R), Machine Learning, Artificial Intelligence

University of Tübingen — Tübingen, Germany
Erasmus+ Exchange Program, Economics — *April 2026 – September 2026*

- **Relevant Coursework:** Quantitative Analysis (R), Time Series Analysis, Monte Carlo Inference, Statistical Inference, MLE, OLS/Logit, Data Science (Python and R)

Skills

Programming Languages: Python, R
Tools: Excel, RStudio, VS Code, Jupyter, Git, GitHub
Libraries: pandas, NumPy, Matplotlib, SciPy, tidyverse
Methods: Data Cleaning, Data Visualization, Regression Analysis, Hypothesis Testing, Time Series Analysis
Machine Learning: Scikit-Learn, XGBoost, LightGBM, TensorFlow, Keras, caret, tidymodels
Languages: Turkish (Native), English (Advanced), German (Beginner)
Certifications: Machine Learning for Time Series Data (DataCamp), Exploratory Data Analysis (DataCamp), Next Generation Banking Simulation (Isbank), Digital Banking (Isbank)

Experience

Istanbul University Informatics Department — Vezneciler, Istanbul
Undergraduate Research Assistant — *November 2023 – June 2024*

- Built a complete database of MIS department heads across all 81 Turkish provinces
- Collected, cleaned, and standardized research datasets in Excel and Google Sheets

Projects

Localify — Slush'D Hackathon
Generative AI, Computer Vision, LLMs, Prompt Engineering — *October 2026*

- Co-built a vision-model pipeline (fal, Apify) that flags which Instagram posts need localization for each market
- Developed prompts for an LLM agent (Manus API) returning per-market change plans as structured JSON; open-sourced

Bitcoin's 21M Cap vs. Inflation Targeting — University of Tübingen
Seminar Paper — *April – August 2026*

- Assessed whether Bitcoin's fixed supply solves the time-inconsistency problem addressed by inflation targeting (Kydland–Prescott, Barro–Gordon, Bernanke–Mishkin), using interwar gold standard evidence
- Concluded that a fixed supply strengthens commitment but is an inferior monetary rule due to debt-deflation risk, price volatility, and a shrinking fee base for network security

Credit Risk and Econometric Modeling in R — University of Tübingen
R, Logit, Monte Carlo Simulation — *August 2026*

- Compared LPM vs. Logit (average partial effects, robust SEs) on 45,000 loan applications, diagnosing quasi-complete separation
- Validated estimator bias/variance/MSE and test power curves via Monte Carlo simulation against asymptotic theory; implemented Bernoulli MLE via numerical optimization
- Modeled China's CO_2 emissions growth with AR models using ADF tests, AIC/BIC lag selection, and Ljung-Box diagnostics

Algorithmic Trading System — Personal Project
Python, XGBoost, Binance API, AWS EC2 — *August – December 2025*

- Trained an XGBoost classifier on RSI, MACD, and OBV indicators to filter trade signals, validated through backtesting
- Implemented hysteresis-based entry/exit logic to hold momentum positions and filter volatility; deployed on AWS EC2

Activities

Young Investors Club — Istanbul University
Researcher — *October 2025 – Present*

- Conduct research and draft analyses supporting the club's social media insights

Data Club — Istanbul University
Vice President — *October 2023 – October 2024*

- Organized an institutional visit to IBB Data Laboratory and managed corporate communications